

14.2 (a) 对 $\|\psi_n\|^2$, $\|\psi_n\|^2 = \int_0^1 \psi_n^2(x) dx = \int_0^1 1 dx = 1$

对 $i \neq j$, 不妨设 $j > i$ 于是 $2^{j+1} = 2^{i+1} 2^{j-i}$

定义区间 $I_m = (\frac{m-1}{2^{j+1}}, \frac{m}{2^{j+1}})$, 不~~难~~知有 $I'_n = (\frac{n-1}{2^{i+1}}, \frac{n}{2^{i+1}})$

在每个 I_n 上, $\psi_i(x) = \text{sgn}(\sin(2^{i+1}\pi x))$ 相等,

有 2^{j-i} 个 I_m 类的区间, 其中 $\int_{I_m} \psi_j(x) dx = - \int_{I_{m+1}} \psi_j(x) dx$

于是 $\langle \psi_i(x), \psi_j(x) \rangle = \int_0^1 \psi_i(x) \overline{\psi_j(x)} dx$

$$= \sum_{n=1}^{2^{j+1}} \int_{I'_n} \psi_i(x) \psi_j(x) dx$$

$$= \sum_{n=1}^{2^{j+1}} \psi_i(x) \int_{I'_n} \psi_j(x) dx = \sum_{n=1}^{2^{j+1}} \psi_i(x) \cdot 0 = 0$$

综上, $\{\psi_n\}_{n=0}^\infty$ 是 $L^2[0,1]$ 中的 ON 系.

(b) 只需证明 $(n_1, k_1) \neq (n_2, k_2)$ 时 $\langle \psi_{n_1, k_1}, \psi_{n_2, k_2} \rangle = 0$

不妨设 $n_2 \geq n_1$,

(1) $n_1 = n_2$ 时, 此时 $\psi_{n_1, k_1}, \psi_{n_2, k_2}$ 的支撑集无交集, 于是 $\langle \psi_{n_1, k_1}, \psi_{n_2, k_2} \rangle = 0$

(2) $n_2 > n_1$ 时, 再分两种情况. ① $\psi_{n_1, k_1}, \psi_{n_2, k_2}$ 的支撑集无交集, $\langle \psi_{n_1, k_1}, \psi_{n_2, k_2} \rangle = 0$

② 若有交集, 则 ψ_{n_2, k_2} 支撑集 I_2 是 ψ_{n_1, k_1} 支撑集的子集.

且在 I_2 上, ψ_{n_1, k_1} 的值一定, 于是

$$\int_0^1 \psi_{n_1, k_1}(x) \psi_{n_2, k_2}(x) dx = \int_{I_1 - I_2} 0 \cdot \psi_{n_1, k_1} dx + \int_{I_2} \psi_{n_1, k_1}(x) \int_{I_2} \psi_{n_2, k_2}(x) dx = 0$$

综上, $\{\psi_{nk} | k=1, 2, \dots, 2^n, n=0, 1, 2, \dots\}$ 是 $L^2[0,1]$ 中的正交系

2. 根据 ON 系的最佳逼近, 有 $S_n(f) = \sum_{k=1}^n \langle f, e_k \rangle e_k$

由于该 ON 系不是 ON 基, 故存在 f , 使得 $f \perp e_k, k=1, 2, 3, \dots$ 且 $f \neq 0$

设 $e_0 = \frac{f}{\|f\|}$, 不难验证 $\{e_k\}_{k=0}^\infty$ 是 H 中的一个正交系

3. 利用 ON 基的等价刻画, 对任意 $f \in H$, 若 $f \perp e_k, k=1, 2, 3, \dots, 0$, 则 $f=0$

于是, 若 $f \perp e_{\delta(k)}, k=1, 2, 3, \dots$ 则 $f \perp e_k, k=1, 2, 3, \dots$, 则 $f=0$

故 $\{e_{\delta(k)}\}_{k=1}^\infty$ 也是 ON 基

4. 若 $\langle f, e_k \rangle = 0, k=1, 2, \dots$ 则 $\|f\|^2 = \langle f, f \rangle = \sum_{k=1}^\infty \langle f, e_k \rangle \overline{\langle f, e_k \rangle} = 0 \therefore f=0$

$\therefore \{e_k\}_{k=1}^\infty$ 是 ON 基

$\Rightarrow$ 若 $\{e_k\}_{k=1}^\infty$ 是 ON 基, 则成立 Parseval 封闭公式

考虑极化恒等式 $\langle f, g \rangle = \frac{1}{4} (\|f+g\|^2 - \|f-g\|^2 + i\|f+ig\|^2 - i\|f-ig\|^2)$

$$= \frac{1}{4} \sum_{k=1}^\infty (\langle f+g, e_k \rangle \overline{\langle f+g, e_k \rangle} - \langle f-g, e_k \rangle \overline{\langle f-g, e_k \rangle} + i \langle f+ig, e_k \rangle \overline{\langle f+ig, e_k \rangle} - i \langle f-ig, e_k \rangle \overline{\langle f-ig, e_k \rangle})$$

$$= \frac{1}{4} \cdot 4 \cdot \sum_{k=1}^\infty \langle f, e_k \rangle \overline{\langle g, e_k \rangle} = \sum_{k=1}^\infty \langle f, e_k \rangle \overline{\langle g, e_k \rangle}$$

14.25 $\Rightarrow$ 反证, 不妨设 $\{\tilde{e}_k\}_{k=1}^{\infty}$ 不是 ON 基,

则存在 $\langle f, \tilde{e}_k \rangle = 0, k=1, 2, \dots$

于是 $\langle f, e_k \rangle = \langle f, e_k - \tilde{e}_k \rangle$

$$\therefore |\langle f, e_k \rangle|^2 = |\langle f, e_k - \tilde{e}_k \rangle|^2 \leq \|f\|^2 \|e_k - \tilde{e}_k\|^2$$

$$\therefore |\langle f, e_k \rangle|^2 = |\langle f, e_k - \tilde{e}_k \rangle|^2 \leq \|f\|^2 \|e_k - \tilde{e}_k\|^2$$

$$\therefore \|f\|^2 = \sum_{k=1}^{\infty} |\langle f, e_k \rangle|^2 \leq \|f\|^2 \sum_{k=1}^{\infty} \|e_k - \tilde{e}_k\|^2 < \|f\|^2$$

矛盾, 于是 $\{e_k\}_{k=1}^{\infty}$ 是 ON 基

$\Leftarrow$ 同理

14.3.1 (1) 利用柯西公式

$$\int_a^b |f_n(x) - f(x)| dx \leq \int_a^b |f_n(x) - f(x)|^2 dx \cdot \int_a^b 1^2 dx = (b-a) \int_a^b |f_n(x) - f(x)|^2 dx$$

∴ 若 $\int_a^b |f_n(x) - f(x)|^2 dx \rightarrow 0 \quad (n \rightarrow \infty)$, 则有 $\int_a^b |f_n(x) - f(x)| dx \rightarrow 0 \quad (n \rightarrow \infty)$

(2) 构造 $f(x) \equiv 0$, $f_n(x) = \sqrt{n} \left(\frac{x-a}{b-a} \right)^{n-1}$

$$\therefore \sup_{n \geq 1} \int_a^b |f_n(x)|^2 dx = \sup_{n \geq 1} \int_a^b n \left(\frac{x-a}{b-a} \right)^{n-1} dx = \sup_{n \geq 1} \left(\frac{x-a}{b-a} \right)^n \Big|_{x=a}^{x=b} = 1 < \infty$$

$$\int_a^b |f_n(x) - f(x)| dx = \sqrt{n} \int_a^b \left(\frac{x-a}{b-a} \right)^{\frac{n-1}{2}} dx = \sqrt{n} \int_0^1 t^{\frac{n-1}{2}} dt \cdot (b-a)^{\frac{3}{2}} = (b-a)^{\frac{3}{2}} \frac{2\sqrt{n}}{n+1} \rightarrow 0 \quad (n \rightarrow \infty)$$

$$\text{但 } \int_a^b |f_n(x) - f(x)|^2 dx = \int_a^b |f_n(x)|^2 dx = 1 \not\rightarrow 0 \quad (n \rightarrow \infty)$$

2. 对任意区间 (A, B) (由于零测集在积分内可忽略, 故只考虑开区间)

可以将区间拆分为 $(A, B) = \bigcup_{k=-n}^n I_k \cap (A, B)$ 其中 $I_k = ((2k-1)\pi, (2k+1)\pi)$, $n = \frac{\max\{|A|, |B|\}}{2\pi} + 1$

于是, 我们只需证 f 在 I_k 上 f 是 Lebesgue 可积

$$\text{而 } \int_{-\pi}^{\pi} f(t) dt = \int_{-\pi}^{\pi} f(t - (2k-1)\pi) dt = \int_{(2k-1)\pi}^{(2k+1)\pi} f(t) dt = \int_{I_k} f(t) dt$$

于是 f 在 I_k 上是 Lebesgue 可积

不妨设 $h = (h-\pi, h+\pi) = (h-\pi, (2k+1)\pi) \cup ((2k-1)\pi, h+\pi)$

$$\begin{aligned} \text{于是 } \int_{-\pi}^{\pi} f(t+h) dt &= \int_{h-\pi}^{h+\pi} f(t) dt = \int_{h-\pi}^{(2k+1)\pi} f(t) dt + \int_{(2k-1)\pi}^{h+\pi} f(t) dt \\ &= \int_{(2k+1)\pi}^{h+\pi} f(t) dt + \int_{h-\pi}^{(2k+3)\pi} f(t) dt = \int_{(2k+1)\pi}^{(2k+3)\pi} f(t) dt = \int_{-\pi}^{\pi} f(t) dt \end{aligned}$$

3. 利用极化恒等式 $\langle f, g \rangle = \frac{1}{4} (\|f+g\|^2 - \|f-g\|^2 + i\|f+ig\|^2 - i\|f-ig\|^2)$

代入 Parseval 封闭公式, 有 $\frac{1}{\pi} \langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx = \frac{a_0 \overline{a_0}}{2} + \sum_{n=1}^{\infty} (a_n \overline{a_n} + b_n \overline{b_n})$

4. 根据定义 $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt$

$$\begin{aligned} \text{于是 } \sum_{n=1}^{\infty} (-1)^{n-1} \frac{b_n}{n} &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n} \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n} \sin(nt) dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \frac{t}{2} dt \end{aligned}$$

对于 x $x = \sum_{n=1}^{\infty} \frac{1}{n} \int_{-\pi}^{\pi} f(t) \sin(nt) dt \sin(nx) = \sum_{n=1}^{\infty} \frac{1}{n} \frac{2\pi (-1)^{n-1}}{n} \sin(nx) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin(nx)$

于是 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n} \sin(nt) = \frac{t}{2}$

$$\therefore \sum_{n=1}^{\infty} (-1)^{n-1} \frac{b_n}{n} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \frac{t}{2} dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) x dx$$

5. 对于 f , 其 Fourier 级数由逐项积分为 $S_n(x) = \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos(kx) + b_k \sin(kx))$

又 $f \in L^2(-\pi, \pi)$ $\therefore \int_{-\pi}^{\pi} |f - S_n|^2 dx \rightarrow 0 \quad (n \rightarrow \infty)$

于是 $\left| \int_E f(x) dx - \int_E (f(x) - S_n(x)) dx \right| \leq \left(\int_E |f - S_n|^2 dx \right)^{1/2} \cdot \left(\int_E 1 dx \right)^{1/2} \leq \left(\int_E |f - S_n|^2 dx \right)^{1/2} \sqrt{2\pi} \rightarrow 0 \quad (n \rightarrow \infty)$

$$\therefore \int_E f(x) dx = \lim_{n \rightarrow \infty} \int_E S_n(x) dx = \frac{a_0}{2} \int_E dx + \sum_{k=1}^{\infty} \int_E (a_k \cos(kx) + b_k \sin(kx)) dx$$

设 $F(x) = \int_0^x f(t) dt$ $\therefore a_0 = 0$ $\therefore a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt = 0$

$$\therefore F(x+2\pi) = \int_0^{x+2\pi} f(t) dt = \int_0^x f(t) dt + \int_x^{x+2\pi} f(t) dt = \int_0^x f(t) dt + \int_{-\pi}^{\pi} f(t) dt = \int_0^x f(t) dt = F(x)$$

$\therefore F$ 是一个周期为 2π 的函数

$$\text{于是 } F(x) = \int_0^x f(t) dt = \sum_{k=1}^{\infty} \left(a_k \int_0^x \cos(kt) dt + b_k \int_0^x \sin(kt) dt \right) = \sum_{k=1}^{\infty} \frac{b_k}{k} + \sum_{k=1}^{\infty} \left(-\frac{b_k}{k} \cos(kx) + \frac{a_k}{k} \sin(kx) \right)$$

6. 不妨设 $a_N f_N(x) = \sum_{k=-N}^N c_k \frac{e^{ikx}}{\sqrt{2\pi}}$

下验证该序列收敛 $N > M$ 时 $\|f_N - f_M\|^2 = \sum_{k=-M}^{-N-1} |c_k|^2 + \sum_{k=M+1}^N |c_k|^2 \rightarrow 0 \quad (M \rightarrow \infty)$

$\therefore \{f_N\}_{N=0}^\infty$ 是一列 Cauchy 列

而 $L^2(-\pi, \pi)$ 是 Hilbert 空间, 于是 $\{f_N\}_{N=0}^\infty$ 收敛 设 $\lim_{N \rightarrow \infty} f_N = f^y g$

对于 g , $\langle g, \frac{e^{ikx}}{\sqrt{2\pi}} \rangle = c_k \therefore \hat{g}(k) = \frac{c_k}{\sqrt{2\pi}} \therefore g$ 是方程组的解

若 h 也满足是一个解, 则 $\langle h - g, \frac{e^{ikx}}{\sqrt{2\pi}} \rangle = 0 \quad k = 0, \pm 1, \pm 2, \dots$

于是 $h = g$, 从而证明了唯一性

7. 不存在, $\{\sin(nx)\}_{n=1}^\infty$ 也是一组正交系, 在该组正交系下,

$$\sum_{n=1}^{\infty} \pi \frac{1}{n} = \pi \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

故该求和范数发散, 从而不存在对应的 $f(x)$ f

14.5.1 求系数 $c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(a-in)x} dx$$

$$= \frac{1}{2\pi(a-in)} (e^{(a-in)\pi} - e^{-(a-in)\pi})$$

$$= \frac{(-1)^n}{2\pi(a-in)} (e^{a\pi} - e^{-a\pi}) = \frac{(-1)^n(a+in)}{2\pi(a^2+n^2)} (e^{a\pi} - e^{-a\pi})$$

于是 $f(x) = e^{ax} = \frac{e^{a\pi} - e^{-a\pi}}{2\pi} \sum_{n=-\infty}^{\infty} \frac{(-1)^n(a+in)}{a^2+n^2} e^{inx}$

此处定义延拓的 f 使 $f(\pi) = f(-\pi) = \frac{e^{a\pi} + e^{-a\pi}}{2}$

验证, 代令 $a=1$, 代 $x=\pi$, 得 $\frac{e^{\pi} + e^{-\pi}}{2} = \frac{e^{\pi} - e^{-\pi}}{2\pi} \sum_{n=-\infty}^{\infty} \frac{(-1)^n(1+in)}{1+n^2} (-1)^n$

于是 $\sum_{n=1}^{\infty} \frac{1}{n^2+1} = \frac{1}{2} \left(\sum_{n=-\infty}^{\infty} \frac{1}{n^2+1} - 1 \right) = \frac{1}{2} \left(\pi \frac{e^{\pi} + e^{-\pi}}{e^{\pi} - e^{-\pi}} - \sum_{n=-\infty}^{\infty} \frac{in}{n^2+1} - 1 \right)$

$$= \frac{\pi}{2} \frac{e^{\pi} + e^{-\pi}}{e^{\pi} - e^{-\pi}} - \frac{1}{2}$$

从而 Fourier 展开正确

2. 先算 $|\sin x|$ 其为偶函数, 计算其系数

$$a_0 = \frac{2}{\pi} \int_0^{\pi} |\sin x| dx = \frac{4}{\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} |\sin x| \cos(2nx) dx = \frac{1}{\pi} \int_0^{\pi} 2 \sin x \cos(2nx) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} (\sin(2n+1)x - \sin(2n-1)x) dx$$

$$= \frac{1}{\pi} \left(\frac{1}{2n+1} - \frac{1}{2n-1} \right) \times 2 = -\frac{4}{(4n^2-1)\pi}$$

于是 $|\sin x| = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(2nx) = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos(2nx)$

$$|\cos x| = |\sin(x + \frac{\pi}{2})| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos(2nx + n\pi)$$

$$= \frac{2}{\pi} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{4n^2-1} \cos(2nx)$$

3. (1) 该函数为偶函数, 计算其系数

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_{2\alpha}^{\pi} (2\alpha - x) dx = \frac{2\alpha}{\pi} (\pi - 2\alpha) = \frac{1}{\pi} (\pi^2 - 4\alpha^2)$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{2\alpha} (2\alpha - x) dx = \frac{4\alpha^2}{\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx = \frac{2}{\pi} \int_0^{2\alpha} (2\alpha - x) \cos(nx) dx$$

$$= \frac{2}{\pi} \frac{2\sin(n\alpha)}{n^2} = \frac{4\sin(n\alpha)}{n^2\pi}$$

即有 $f(x) = \frac{2\alpha^2}{\pi} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin^2(n\alpha)}{n^2} \cos(nx)$

(2) 一致收敛性: 对 $0 < 2\alpha < \pi$, 有 $\int_0^{\pi} \left| \frac{f(x+t) + f(x-t)}{2} - f(x) \right| \frac{dt}{t} \leq \int_0^{\delta} dt = \delta < \infty$

而延拓后的 f 处处连续, 故 Fourier 函数一致收敛

(2) 令 $\alpha=1$, $x=0$, 则 $f(x) = \frac{2}{\pi} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin^2 n}{n^2} = 2$

$$\therefore \sum_{n=1}^{\infty} \frac{\sin^2 n}{n^2} = \pi \frac{\pi-1}{2}$$

$$\sum_{n=1}^{\infty} \frac{\cos n}{n^2} = \sum_{n=1}^{\infty} \frac{1 - \sin^2 n}{n^2} = \frac{\pi^2}{6} - \frac{\pi-1}{2}$$

4. 方法一: 法对 $f(x) = x^4$, 进行 Fourier 展开

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^4 dx = \frac{2\pi^4}{5} \quad a_n = \frac{2}{\pi} \int_0^{\pi} x^4 \cos(nx) dx = (-1)^{n-1} \frac{8\pi^2 n^2 + 48}{n^4}$$

提 $f(x) = x^4 = \frac{\pi^4}{5} + \sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{8\pi^4}{n^4} + \frac{48}{n^4} \right) \cos(n\pi)$

代入 $x = \pi$, 有 $\pi^4 = \frac{\pi^4}{5} - \sum_{n=1}^{\infty} \left(\frac{8\pi^4}{n^4} + \frac{48}{n^4} \right)$

从而 $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$

方法二: 用指数形式, 展开 $f(x) = x^2$, 有

$$f(x) = x^2 = \frac{\pi^2}{3} + 2 \sum_{k=1}^{\infty} \frac{(-1)^k}{k^2} e^{ikx}$$

$\therefore \frac{1}{2\pi} \|f(x)\|^2 = \frac{\pi^4}{9} + 2 \sum_{n=1}^{\infty} \frac{4}{n^4}$ (Parseval 公式)

另一方面 $\frac{1}{2\pi} \|f(x)\|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^4 dx = \frac{\pi^4}{5}$

于是 $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{8} \left(\frac{\pi^4}{5} - \frac{\pi^4}{9} \right) = \frac{\pi^4}{90}$

5. $\forall \delta \in (0, \pi) \quad |t| \in (\delta, \pi)$ 时, $\sin^2(\frac{\pi}{2}t) \geq \sin^2(\frac{\delta}{2}) > 0$

于是 $F_n(t) = \frac{1}{n} \frac{\sin^2(\frac{\pi}{2}t)}{\sin^2(\frac{\delta}{2})} \leq \frac{1}{n \cdot \sin^2(\frac{\delta}{2})}$ 设 $f(x)$ 在 $[-\pi, \pi]$ 上最大值为 M

于是 $\lim_{n \rightarrow \infty} \int_{\delta < |t| \leq \pi} F_n(t) dt \leq \lim_{n \rightarrow \infty} \int_{\delta < |t| \leq \pi} \frac{1}{n \cdot \sin^2(\frac{\delta}{2})} dt \leq \lim_{n \rightarrow \infty} \frac{2\pi}{n \cdot \sin^2(\frac{\delta}{2})} = 0$

$\therefore \lim_{n \rightarrow \infty} G_n(f)(x_0) - f(x_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(x_0-t) - f(x_0)] F_n(x_0, t) dt$

$= \frac{1}{2\pi} \left(\int_{|t| < \delta} [f(x_0-t) - f(x_0)] F_n(t) dt + \int_{\delta < |t| \leq \pi} [f(x_0-t) - f(x_0)] F_n(t) dt \right)$

$\leq \frac{M}{2\pi} \int_{-\pi}^{\pi} F_n(t) dt + \int_{\delta}$

由 δ 取 δ' 使 $|f(x_0-t) - f(x_0)| < \frac{\varepsilon}{2}$, 设 $f(x)$ 在 $[-\pi, \pi]$ 上最大值为 M ,

设对 δ' , $n > N$ 时, $\int_{\delta' < |t| \leq \pi} F_n(t) dt < \frac{\varepsilon}{4M}$

于是 $n > N$, δ 取 δ' 时, $G_n(f)(x_0) - f(x_0) \leq \frac{1}{2\pi} \frac{\varepsilon}{4M} \int_{-\pi}^{\pi} F_n(t) dt + 2M \int_{\delta' < |t| \leq \pi} F_n(t) dt$

$= \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$

所以 $\lim_{n \rightarrow \infty} (G_n(f)(x_0) - f(x_0)) = 0$

$\therefore \lim_{n \rightarrow \infty} G_n(f)(x_0) = f(x_0)$

6. 若 $\lim_{n \rightarrow \infty} \lim_{x \rightarrow x_0} S_n(x) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{ikx_0} = f(x_0) A$

$\therefore \lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{k=0}^n S_n(x_0) = \lim_{n \rightarrow \infty} G_n(f) = \lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{k=0}^n S_n(x_0) = \lim_{n \rightarrow \infty} S_n(x_0) = A$

根据上一问 $A = f(x_0)$

7. (1) 将 f 的延拓函数 $F(x) = \begin{cases} f(x) & x \in [0, \pi) \\ f(-x) & x \in [-\pi, 0) \end{cases}$

不难验证 F 在 $[-\pi, \pi]$ 上连续, 且在 $[-\pi, 0)$ $(0, \pi]$ 上一致连续光滑

利用 14.5.7 命题, 即有该余弦级数一致收敛

(2) 必要性: 设 G 为 F 的正弦级数, 定义延拓函数为 $F(x) = \begin{cases} f(x) & x \in [0, \pi] \\ -f(-x) & x \in [-\pi, 0) \end{cases}$

必要性: 若 $G(x)$ 一致收敛到 $f(x)$, 则 G 在 $[-\pi, \pi]$ 上一致收敛到 F

所以 $F(0^+) = F(0^-) = \frac{F(0^+) + F(0^-)}{2} = 0$ $F(-\pi) = F(\pi^-) = \frac{F(-\pi) + F(\pi^-)}{2} = 0$

充分性: 若 $f(0) = f(\pi) = 0$, 则 F 在 $[-\pi, \pi]$ 上是 C^1 的 且 $F(-\pi) = F(\pi)$

从而利用命题 14.5.7, 即有 G 收敛到 F

$$8, \text{ 令 } t = x^p, \text{ 则 } \int_0^1 (1-x^p)^{-\frac{1}{p}} = \int_0^1 (1-t)^{-\frac{1}{p}} t^{\frac{1}{p}-1} dt$$

$$= B(1-\frac{1}{p}, \frac{1}{p}) = \frac{\Gamma(1-\frac{1}{p})\Gamma(\frac{1}{p})}{\Gamma(1)} =$$

$$= \frac{\Gamma(1-\frac{1}{p})\Gamma(\frac{1}{p})}{\sin(\frac{\pi}{p})} = \frac{\pi}{\sin(\frac{\pi}{p})}$$

$$\text{令 } x^p = t, \text{ 则 } \int_0^\infty \frac{x^{a-1}}{1+x^b} dx = \int_0^\infty \frac{t^{\frac{a}{p}-1}}{1+t} \frac{1}{p} dt = \frac{1}{p} \int_0^\infty \frac{t^{\frac{a}{p}-1}}{1+t} dt = \frac{\pi}{b \sin(\frac{a\pi}{p})}$$